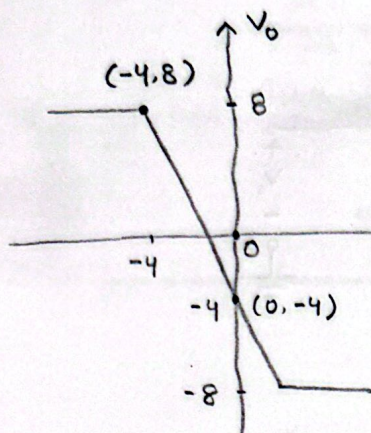


Answer to the Question No-1

a) Points are, (V_{in}, V_o) , $(-4.8, 8)$, $(0, -4)$

$$\frac{V_{in} + 4}{-4 - 0} = \frac{V_o - 8}{8 + 4}$$

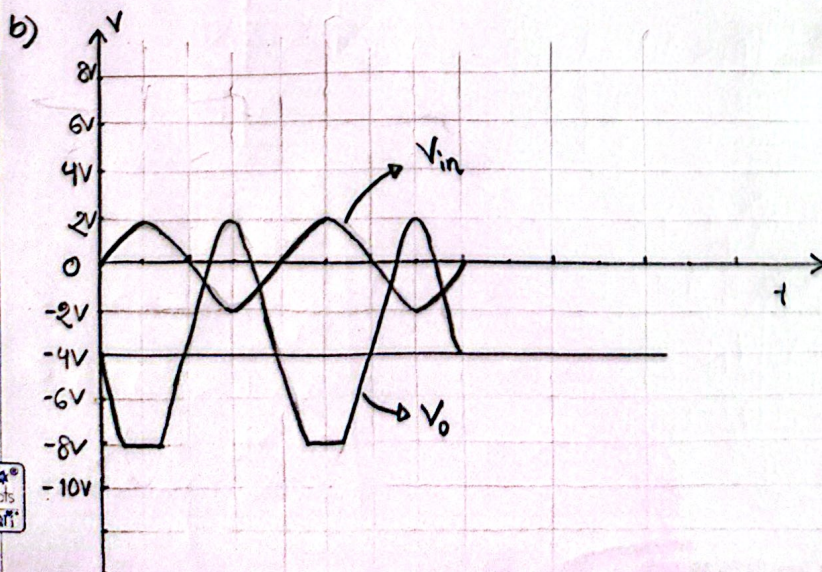
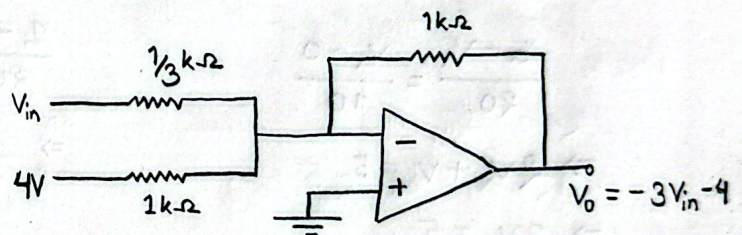
$$\Rightarrow -4V_o + 32 = 12V_{in} + 48$$

$$\Rightarrow 4V_o = -12V_{in} - 48 + 32$$

$$\Rightarrow V_o = -3V_{in} - 4$$

$$\frac{R_f}{R_i} = 3$$

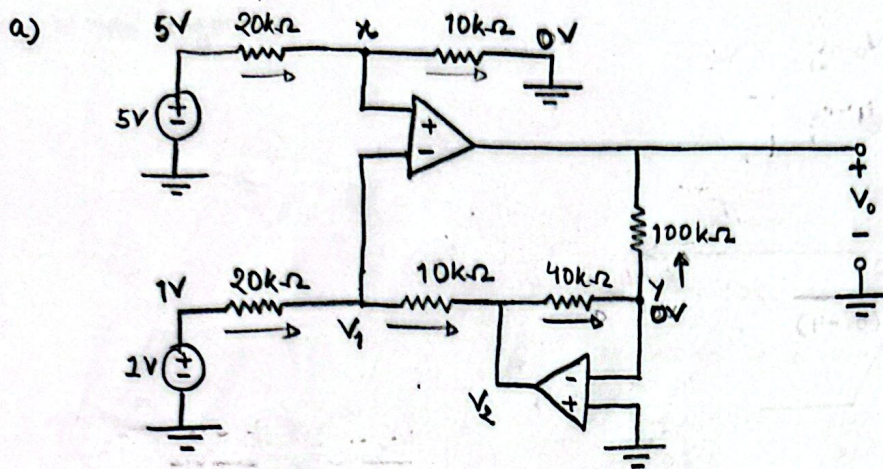
$$R_f = 1k\Omega; R_i = \frac{1}{3}k\Omega$$



$$V_{in} = 2 \sin(\omega t)$$

$$V_o = -3V_{in} - 4$$

$$V_o = -6 \sin(\omega t) - 4$$

Answer to the Question No-2On x point,

$$\frac{5 - V_x}{20} = \frac{V_x - 0}{10}$$

$$\Rightarrow 2V_x + V_x = 5$$

$$\Rightarrow 3V_x = 5$$

$$\Rightarrow V_x = \frac{5}{3} \text{ V}$$

$$V_1 = V_x$$

$$\therefore V_1 = \frac{5}{3} \text{ V}$$

On V_1 ,

$$\frac{1 - V_1}{20} = \frac{V_1 - V_2}{10}$$

$$\Rightarrow \frac{1 - \frac{5}{3}}{20} = \frac{\frac{5}{3} - V_2}{10}$$

$$\Rightarrow -\frac{1}{3} = \frac{5}{3} - V_2$$

$$\Rightarrow V_2 = 2 \text{ V}$$

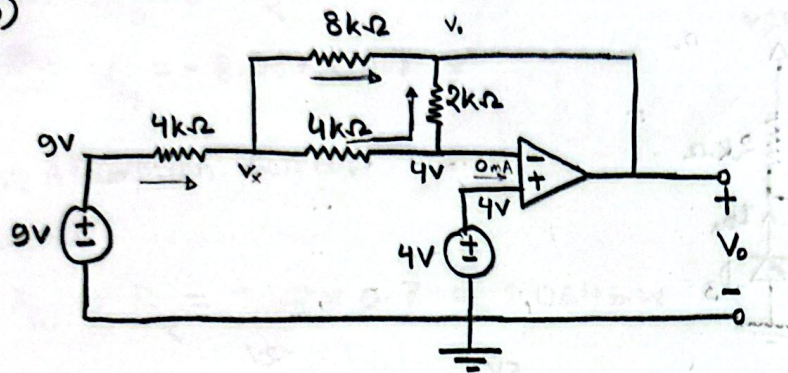
On y point

$$\frac{V_2 - 0}{40} = \frac{0 - V_o}{100}$$

$$\Rightarrow \frac{2 - 0}{40} = \frac{-V_o}{100}$$

$$\Rightarrow V_o = -5 \text{ V}$$

b)

For V_x ,

$$\Rightarrow \frac{V_x - 9}{4} + \frac{V_x - V_0}{8} + \frac{V_x - 4}{4} = 0$$

$$\Rightarrow 18 - 2V_x - V_x + V_0 - 2V_x + 8 = 0$$

$$\Rightarrow V_0 - 5V_x = -26 \quad \text{--- (1)}$$

Again,

$$\frac{V_x - 4}{4} = \frac{4 - V_0}{2}$$

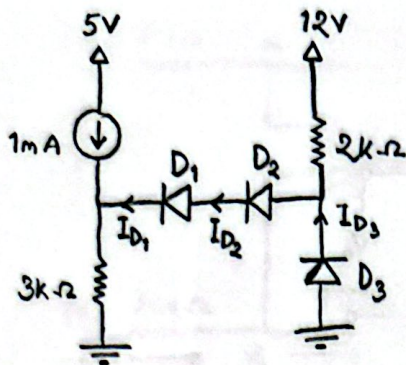
$$\Rightarrow V_x - 4 - 8 + 2V_0 = 0$$

$$\Rightarrow 2V_0 + V_x = 12 \quad \text{--- (2)}$$

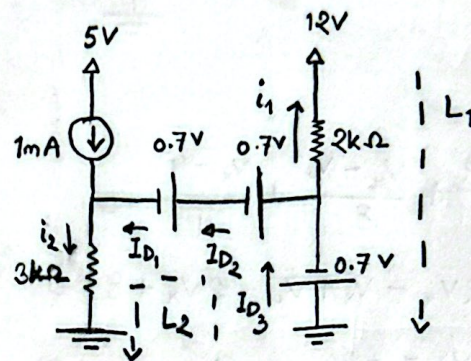
Solving (1) and (2)

$$V_0 = \frac{34}{11} \text{ V}, \quad V_x = \frac{64}{11} \text{ V}$$

(Answer)

Answer to the Question No-3

Assume, $D_1 \rightarrow \text{F.B.}; I_{D1} > 0$
 $D_2 \rightarrow \text{F.B.}; I_{D2} > 0$
 $D_3 \rightarrow \text{F.B.}; I_{D3} > 0$



Line 1,

$$12 - 0 = -2i_1 - 0.7$$

$$\Rightarrow 12 + 0.7 = -2(I_{D3} - I_{D2})$$

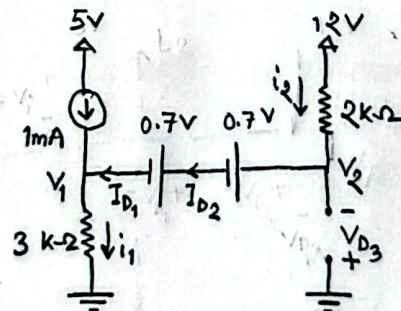
$$\Rightarrow I_{D2} - I_{D3} = 6.35$$

Here, $I_{D1} = I_{D2}$
 $\Rightarrow I_{D1} - I_{D2} = 0$

Line 2, $0 - 0 = 0.7 + 0.7 + 0.7 + 3i_2$
 $\Rightarrow I_{D1} + 1 = -0.7$
 $\Rightarrow I_{D1} = -1.7 \text{ mA} \neq 0$

$\therefore$ Wrong Assumption.

Again, Assume, $D_1 \rightarrow \text{F.B.}; I_{D1} > 0$
 $D_2 \rightarrow \text{F.B.}; I_{D2} > 0$
 $D_3 \rightarrow \text{R.B.}; V_{D3} < 0.7$



Supernode V_1, V_2 ,

$$\frac{V_1 - 0}{3} + \frac{V_2 - 12}{2} - 1 = 0$$

$$\Rightarrow 2V_1 + 3V_2 = 42 \quad \text{--- (I)}$$

$$-V_1 + V_2 = 0.7 + 0.7$$

$$\Rightarrow V_1 - V_2 = -1.4 \quad \text{--- (II)}$$

Solving (I) and (II),

$$V_1 = 7.56 \text{ V}, V_2 = 8.96 \text{ V}$$

$$i_1 = \frac{7.56}{3} = 2.52 \text{ mA}$$

$$i_2 = \frac{8.96}{2} = 1.52 \text{ mA}$$

$$I_{D1} = I_{D2} = i_2$$

$$I_{D1} = 1.52 \text{ mA} > 0 \quad \checkmark$$

$$I_{D2} = 1.52 \text{ mA} > 0 \quad \checkmark$$

$$V_{D3} = V_2 - V_1 = 8.96 - 7.56 = 1.4 \text{ V}$$

Now, $V_{D3} = 0 - V_2$

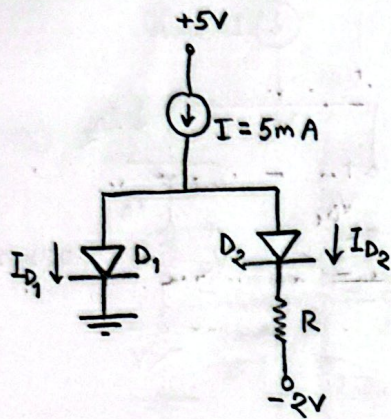
$$V_{D3} = -8.96V < 0.7 \checkmark$$

$\therefore$ Assumption Correct.

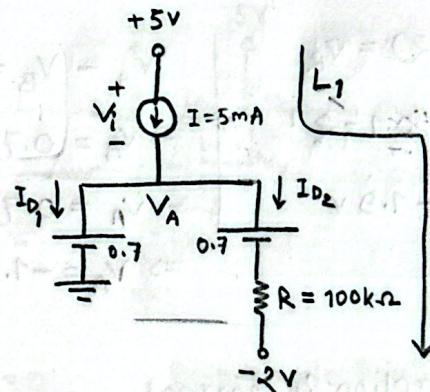
$$P_{D1} = P_{D2} = 1.52 \times 0.7 = 1.064 \text{ mW}$$

$$P_{D3} = 0 \text{ mW}$$

Answer to the Question No-4

CVD Model with $V_{D_0} = 0.7V$

- a) Assume, D_1 is on. $I_{D_1} > 0$
 D_2 is on. $I_{D_2} > 0$



Here, .

$$V_A = 0.7V$$

Line 1,

$$5 + 2 = V_i + 0.7 + I_{D_2} R$$

$$V_i = 5 - 0.7 = 4.3V$$

$$\Rightarrow 7 = 4.3 + 0.7 + 100 I_{D_2}$$

$$\Rightarrow I_{D_2} = 0.02 \text{ mA} > 0 \checkmark$$

$$I_{D_1} = I - I_{D_2}$$

$$= 5 - 0.02$$

$$= 4.98 \text{ mA} > 0 \checkmark$$

$\therefore$ Assumption is correct.

b) Assume, D_1 is off. $V_{D1} < 0.7$

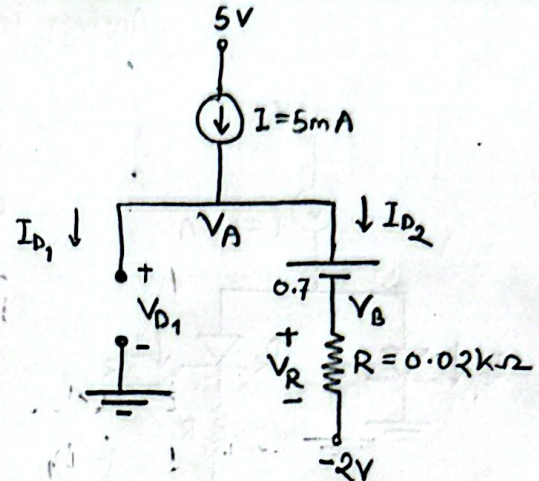
D_2 is on. $I_{D2} > 0$

$$I_{D1} = 0 \text{ mA}, I_{D2} = I = 5 \text{ mA} > 0 \checkmark$$

$$V_R = I_{D2} \times R = 5 \times 0.02 = 0.1 \text{ V}$$

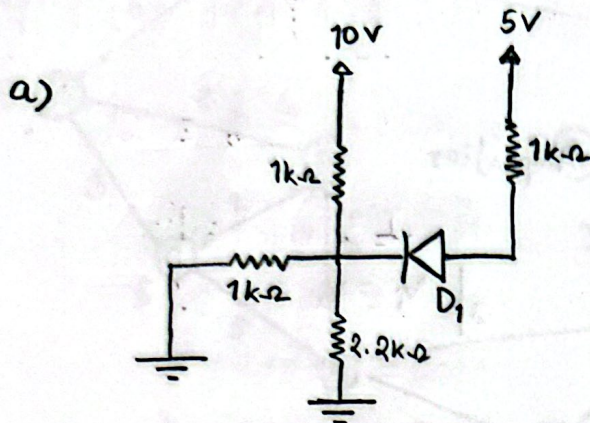
$$\begin{array}{l|l} V_B - (-2) = V_R & V_A - V_B = 0.7 \\ \Rightarrow V_B = 0.1 - 2 & \Rightarrow V_A = 0.7 + V_B \\ \Rightarrow V_B = -1.9 \text{ V} & \Rightarrow V_A = 0.7 - 1.9 \\ & \Rightarrow V_A = -1.2 \text{ V} \end{array}$$

$$V_{D1} = V_A = -1.2 \text{ V} < 0.7 \checkmark$$



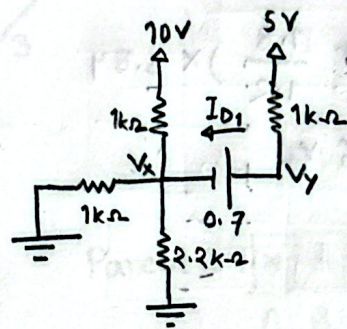
$\therefore$ Assumption is correct.

Answer to the Question No-5



Assume, D_1 is On. $I_{D1} > 0$

Supernode, V_x, V_y ,



$$\frac{V_x - 10}{1} + \frac{V_x - 0}{1} + \frac{V_x - 0}{2.2} + \frac{V_y - 5}{1} = 0$$

Super node Relation,

$$-V_y - V_x = 0.7$$

$$\Rightarrow 11V_x - 110 + 11V_x + 5V_x + 11V_y - 55 = 0$$

$$\Rightarrow V_x - V_y = -0.7 \quad \text{--- (ii)}$$

$$\Rightarrow 27V_x + 11V_y = 165 \quad \text{--- (i)}$$

Solving (i) and (ii),

$$V_x = \frac{1727}{380} = 4.54 \text{ V}$$

$$V_y = \frac{1461}{380} = 3.84 \text{ V}$$

$$I_{D1} = \frac{5 - 3.84}{1} = 1.16 \text{ mA} > 0 \checkmark$$

Assumption is correct.

$\therefore$ Diode (D_1) is on.

$$b) V_a = V_y$$

$$\Rightarrow V_a = 3.84V$$

The figure is of Non-Inverting Amplifier

$$V_o = \left(1 + \frac{R_f}{R_1}\right) \times V_i$$

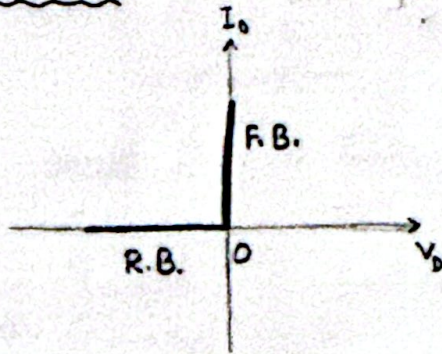
$$| V_i = V_a$$

$$\Rightarrow V_o = \left(1 + \frac{1.8}{1.2}\right) \times 3.84$$

$$\Rightarrow V_o = 9.6V$$

Answer to the Question No-6

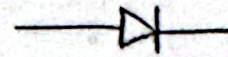
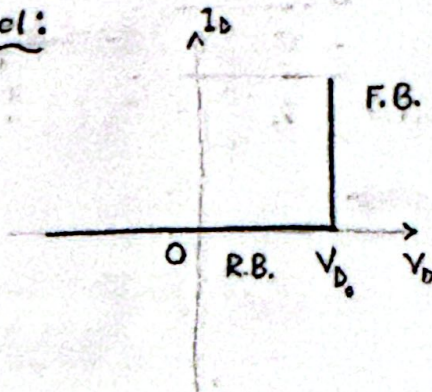
Ideal Model:



F.B. $\rightarrow$ $\rightarrow V_D = 0; I_D > 0$
Short circuit

R.B. $\rightarrow$ $\rightarrow V_D < 0; I_D = 0$
open circuit

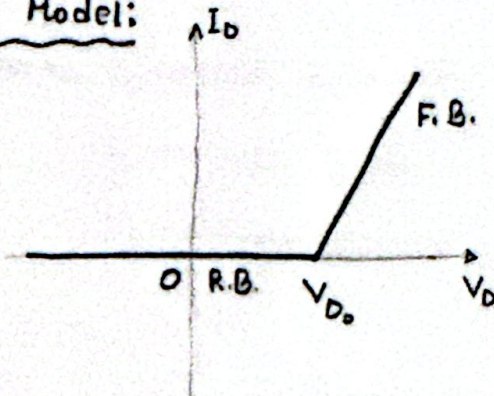
CVD Model:



F.B. $\rightarrow$ $\rightarrow V_D = V_{D0}; I_D > 0$
Voltage source

R.B. $\rightarrow$ $\rightarrow V_D < V_{D0}; I_D = 0$
open circuit

CVD + R Model:



F.B. $\rightarrow$ $\rightarrow V_D = V_{D0} + I_D R_s$
Voltage source + Resistance in series
 $I_D > 0$

R.B. $\rightarrow$ $\rightarrow V_D < V_{D0}; I_D = 0$
open circuit